

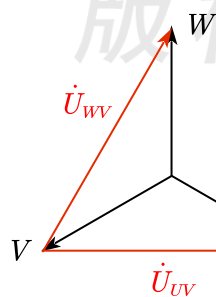
电路理论 mooc 答案——第十五章

仅限上海交通大学电路理论学科营使用

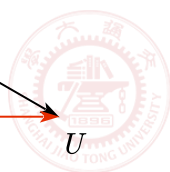
解析

请勿用于其他用途

1.解:



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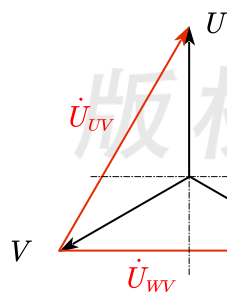
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选C

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2.解:电源相序为负序

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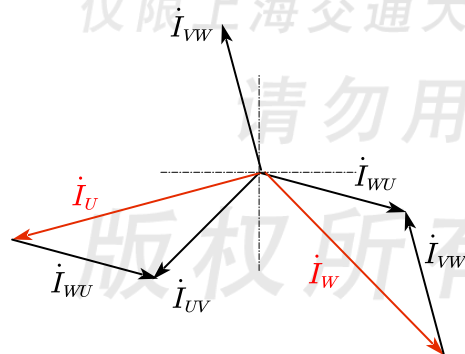
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选D

3.解:

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选A

4.解: 仅限上海交通大学电路理论学科营使用

Let $\dot{U}_U = 220 \angle 0^\circ \text{V}$, then $\dot{U}_{UY} = \dot{U}_U + \dot{U}_V = 220 \angle 0^\circ + 220 \angle -120^\circ = 220 \angle -90^\circ \text{V}$.

选B

5.解: 版权所有 翻录必究

Let $\dot{U}_U = 220 \angle 0^\circ \text{V}$,

then $\dot{U}_{UW} = \dot{U}_U + \dot{U}_V - \dot{U}_W = 220 \angle 0^\circ + 220 \angle -120^\circ - 220 \angle 120^\circ = 440 \angle -90^\circ \text{V}$.

选D

6.解:

$T - \Pi$ 变换后, $\dot{I}_A = \frac{\dot{U}_A}{Z_l + \frac{1}{3}Z} = \frac{220}{1 + j + \frac{1}{3}(12 + 12j)} = 22\sqrt{2} \angle -45^\circ \text{A}$

$\dot{U}_{A'} = \dot{U}_A \cdot \frac{\frac{1}{3}Z}{Z_l + \frac{1}{3}Z} = \frac{4}{5}\dot{U}_A = 176 \angle 0^\circ \text{V}$

对称负载, $U_L = \sqrt{3}U_P = 176\sqrt{3}$, $I_L = \frac{1}{\sqrt{3}}I_P = \frac{22}{3}\sqrt{6} \text{A}$

选C

7.解:

右侧直接分别T型去耦, 最后只剩下三个对称的无耦合电感, $U_{Right} = 0$.

$\dot{U}_{Left} = \frac{1}{2}(\dot{U}_A + \dot{U}_B) = \frac{1}{2}(220 \angle 0^\circ + 220 \angle -120^\circ) = 110 \angle -60^\circ \text{V}$

选B

8.解:

均为对称负载, $\dot{U}_{Left} = \dot{U}_{Right} = 0$, R上始终电压为0.

选A

9.解:

A. $\dot{I}_{A1} = \dot{I}_R + \dot{I}_L$

B. $\dot{U}_{BN} = const$

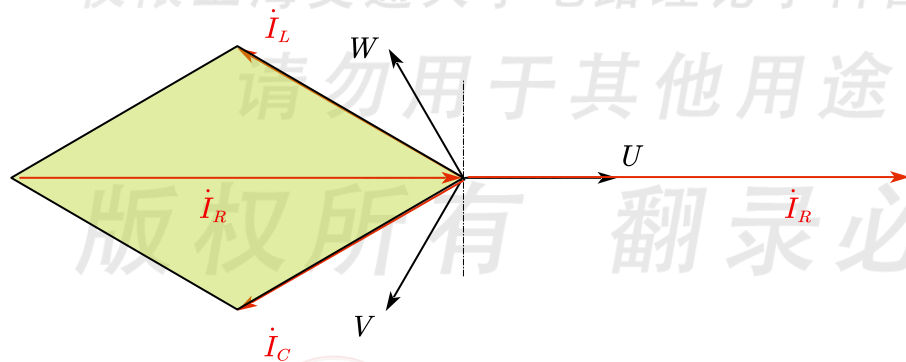
C. $|\dot{I}_{A2}| = |\dot{I}_{A3}| = \frac{U_P}{R}$

D. 对称负载 $\dot{I}_{A0} = 0$, 接入后, $|\dot{I}_{A0}| = |\dot{I}_L| > 0$

选D

10.解:

Let $\dot{U}_U = 10 \angle 0^\circ V$, then $\dot{I} = \dot{I}_R + \dot{I}_L + \dot{I}_C = 0$.



$$\begin{cases} \frac{U_P}{\omega L} = \omega C U_P \\ \sqrt{3} \omega C U_P = \frac{U_P}{R} \end{cases} \Rightarrow \begin{cases} \omega = \frac{1}{\sqrt{LC}} \\ \sqrt{3} \omega R C = \sqrt{3} R \sqrt{\frac{C}{L}} = 1 \end{cases}$$

选B

11.解:

Let $\dot{U}_U = 220 \angle 0^\circ V$, then $3U_P I_P \cos \varphi = P_1 \Rightarrow I_P = 10 A$, $\dot{I}_{U1} = I_P \angle \varphi = 10 \angle -30^\circ A$.

$\dot{I}_{U2} = \frac{\dot{U}_U}{Z} = 10 \angle 30^\circ A$, then $\dot{I}_U = \dot{I}_{U1} + \dot{I}_{U2} = 10\sqrt{3} \angle 0^\circ A$, $\varphi' = \angle \left(\frac{\dot{U}_U}{\dot{I}_U} \right) = 0^\circ$

选C

12.解:

电阻性负载功率与相角无关

选A

13.解:

Let $\dot{U}_U = 220 \angle 0^\circ \text{V}$, $\dot{I}_U = 1 \angle \varphi \text{A}$

$$P_W = \operatorname{Re}\{\dot{U}_{UV} \dot{I}_V^*\} = \operatorname{Re}\{220\sqrt{3} \angle -30^\circ \cdot 1 \angle (-\varphi + 120^\circ)\} = 220\sqrt{3} \sin \varphi = 132\sqrt{3} \text{W}$$

选C

14.解:

$T - \Pi$ 变换后, $Z_{eq} = Z_l + Z_1 // \left(\frac{1}{3} Z_2\right) = (2 + j2) \Omega$

Let $\dot{U}_U = 220 \angle 0^\circ \text{V}$, then $\dot{I}_U = \frac{\dot{U}_U}{Z_{eq}} = (55 - 55j) \text{A}$.

$$P_W = \operatorname{Re}\{\dot{U}_{UV} \dot{I}_U^*\} = \operatorname{Re}\{220\sqrt{3} \angle 30^\circ \cdot (55 + 55j)\} = 7671 \text{W}$$

选D

Remark: 这里是380和 $220\sqrt{3}$ 的精度问题, 做题用哪个都可以。

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